Number System

25/04/2026

* Binary Number System

- Binary digits (0 & 1) are called bits → Smallest unit of data.
- eg: $(1\ 0\ 1\ 0\ 1)_2$ ⇒ $(21)_{10}$

  $2^4\ 2^3\ 2^2\ 2^1\ 2^0$ ⇒ $1\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

  $= 16 + 4 + 1 = 21$

- 10101.11 (2^{-1}, 2^{-2})

$1\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 1\times2^{-2}$

$16 + 4 + 1 +$

MSB → Most significant bit LSB → Least significant bit

eg: 10101 ($b_4b_3b_2b_1b_0$) = 21

MSB LSB replace by 0

10100 = 20 (diff 1)

00101 = 5 (diff 16)

1 Nibble = 4 bits. (represent BCD & Hexadecimal no)

1 Byte = 8 bits

1 Word = 16 bits = 2 bytes

1 double word = 32 bits = 4 bytes.

* Conversion from Decimal

- To binary ⇒ $(126)_{10} = (?)_2$

2	126	
2	63	→ 0
2	31	→ 1
2	15	→ 1
2	7	→ 1
2	3	→ 1
	1	→ 1

⇒ $(1111110)_2$

- To Octal

$(126)_{10} \Rightarrow (\ )_8$

$$\begin{array}{r|l} 8 & 126 \\ \hline 8 & 15 \rightarrow 6 \\ \hline 8 & 1 \rightarrow 7 \\ \hline & 0 \rightarrow 1 \end{array} \Rightarrow (176)_8$$

- To Hexadecimal

$(126)_{10} \Rightarrow (\ )_{16}$

$$\begin{array}{r|l} 16 & 126 \\ \hline 16 & 7 \rightarrow 14 \rightarrow E \\ \hline & 0 \rightarrow 7 \rightarrow 7 \end{array} \Rightarrow (7E)_{16}$$

- To convert to octal we can use binary group. 3 bit

$(1111110)_2$

$$\begin{array}{ccc} 001 & 111 & 110 \\ \downarrow & & \downarrow \\ 1 & 7 & 6 \end{array} \Rightarrow (176)$$

- 111 by → hexadecimal (4 group)

$(1111110)_2 \Rightarrow$

$$\begin{array}{cc} 0111 & 1110 \\ \downarrow & \downarrow \\ 7 & E \end{array} \quad (14)$$

$(7E)_{16}$

- From hexa → binary

$(7E)_{16}$

$$\begin{array}{cc} \downarrow & \downarrow \\ 0111 & 1110 \end{array} \rightarrow (1111110)_2$$

- for hexa ⇒ 0 - 9 → same

10 - 15 →

10 - A, 11 - B, 12 - C, 13 - D, 14 - E, 15 - F

eg: 1. $(1010)_2 \rightarrow$ Decimal

$1\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0 = 8 + 2 = 10$

2. $(1111)_2 \rightarrow$ Decimal

$1\times2^3 + 1\times2^2 + 1\times2^1 + 1\times2^0 = 8 + 4 + 2 + 1 = 15$

3. $(18)_{10} \rightarrow$ binary

$$\begin{array}{r|l} 2 & 18 \\ 2 & 9 \rightarrow 0 \\ 2 & 4 \rightarrow 1 \\ 2 & 2 \rightarrow 0 \\ & 1 \rightarrow 0 \end{array}$$

~~10010~~ $\Rightarrow (10010)_2$

4. $(32)_{10} \rightarrow$ binary

$$\begin{array}{r|l} 2 & 32 \\ 2 & 16 \rightarrow 0 \\ 2 & 8 \rightarrow 0 \\ 2 & 4 \rightarrow 0 \\ 2 & 2 \rightarrow 0 \\ & 1 \rightarrow 0 \end{array}$$

~~100000~~ $(100000)_2$

$2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$

16

5. $(10011)_2 \rightarrow$ Decimal

$1\times2^4 + 0\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 = 16 + 2 + 1 = 19$

6. $(27)_{10} \rightarrow$ binary

$$\begin{array}{r|l} 2 & 27 \\ 2 & 13 \rightarrow 1 \\ 2 & 6 \rightarrow 1 \\ 2 & 3 \rightarrow 0 \\ & 1 \rightarrow 1 \end{array}$$

$(11011)_2$

7. $(11100)_2 \rightarrow$ Decimal

$1\times2^4 + 1\times2^3 + 1\times2^2 = 16 + 8 + 4 = 28$

8. $(A)_{16} \rightarrow$ binary

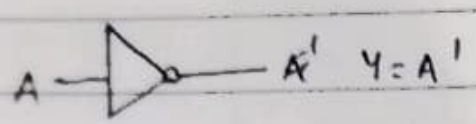
$10 \rightarrow$

$$\begin{array}{r|l} 2 & 10 \\ 2 & 5 \rightarrow 0 \\ 2 & 2 \rightarrow 1 \\ & 1 \rightarrow 0 \end{array}$$

$\Rightarrow 1010$

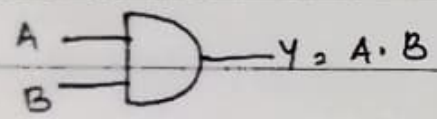
Logic Gates

1) NOT



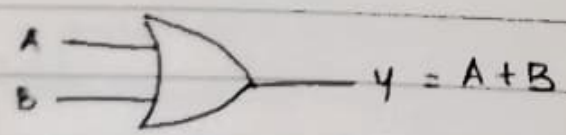
A	Y
0	1
1	0

2) AND



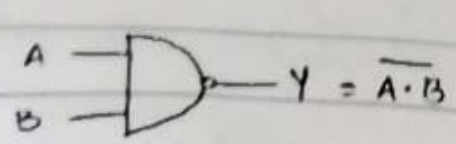
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

3) OR



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

4) NAND



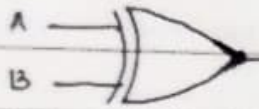
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

5) NOR

$Y = \overline{A+B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6) XOR

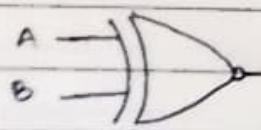


$Y = A \oplus B$

$= A\bar{B} + \bar{A}B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7) XNOR



$Y = \overline{A \oplus B}$

$AB + \bar{A}\bar{B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1